

2.5.5 Configuration Register Settings

For the measurement, set the register values:

- Select multiplexer settings for AINP and AINN to measure leads 1 and 2 of the RTD
- Enable the PGA, set gain to desired value
- Select data rate and digital filter settings
- Select reference input to measure R_{REF} for ratiometric measurement
- Enable the internal reference (the IDAC requires an enabled internal reference)
- Set IDAC magnitude and select IDAC1 output pin to drive the reference resistor R_{REF} and select IDAC2 output pin to drive lead 2 of the RTD

2.6 Four-Wire RTD Measurement, Low-Side Reference

A four-wire RTD has the best measurement accuracy. The lead resistance reacting with the IDAC current does not add an error term for the ADC measurement.

2.6.1 Schematic

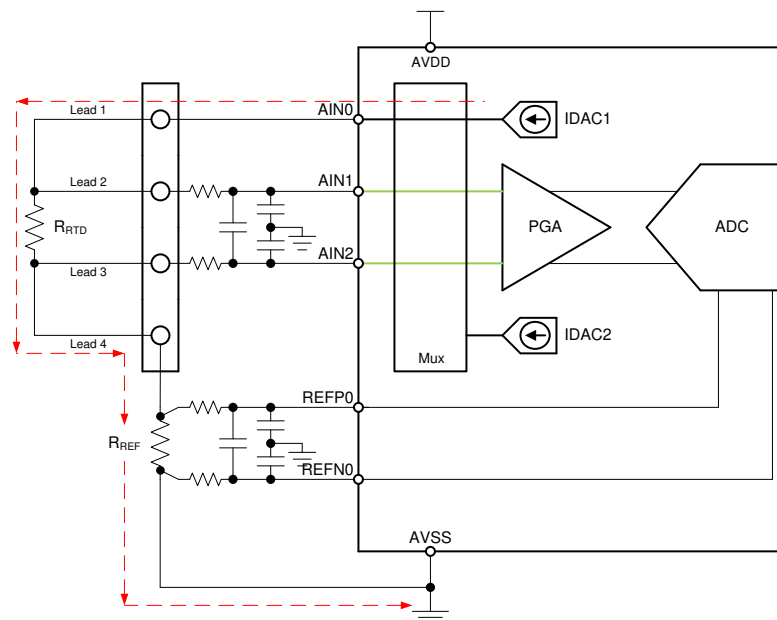


Figure 2-6. Four-Wire RTD, Low-Side Reference Measurement Circuit

2.6.2 Pros and Cons

Pros:

- Single IDAC current is used for sensor and reference resistor excitation
- Most accurate RTD measurement, no need for lead compensation
- Ratiometric measurement, IDAC noise and drift are cancelled

Cons:

- The four-wire RTD is the most expensive RTD configuration

2.6.3 Design Notes

This design should be exactly the same as in the two-wire RTD design in [Section 2.1](#). The IDAC current is being routed through an alternate input and measurements are taken from AIN1 and AIN2. However, the considerations in reference resistor size, IDAC current, reference voltage, and PGA input voltage are exactly the same.

The measurement circuit requires:

- Single dedicated IDAC output pin
- AINP and AINN inputs
- External reference input

- Precision reference resistor

To verify that the design is within the ADC range of operation, Calculate the voltages for AIN1 and AIN2 and the maximum differential input voltage. Verify that V_{AIN1} and V_{AIN2} are within the input range of the PGA given the gain setting and supply voltage. Use the maximum RTD resistance based on the desired temperature measurement.

$$V_{AIN1} = I_{IDAC1} \cdot (R_{RTD} + R_{REF}) \quad (65)$$

$$V_{AIN2} = I_{IDAC1} \cdot R_{REF} \quad (66)$$

Additionally, verify that the voltage seen at the IDAC pin (where $V_{AIN0} = V_{AIN1}$) is within the current source compliance voltage. When the IDAC output voltage rises too close to AVDD, the IDAC loses compliance and the excitation current is reduced.

The reference resistor, R_{REF} must be a precision resistor with high accuracy and low drift. Any error in the R_{REF} reflects the same error in the RTD measurement. The REFP0 and REFN0 pins are shown connecting to the R_{REF} resistor as a Kelvin connection to get the best measurement of the reference voltage. This eliminates any series resistance as an error from the reference resistance measurement.

2.6.4 Measurement Conversion

$$\text{Output code} = 2^{23} \cdot \text{Gain} \cdot V_{RTD} / V_{REF} = 2^{23} \cdot \text{Gain} \cdot (I_{IDAC1} \cdot R_{RTD}) / (I_{IDAC1} \cdot R_{REF}) = 2^{23} \cdot \text{Gain} \cdot R_{RTD} / R_{REF} \quad (67)$$

$$R_{RTD} = R_{REF} \cdot \text{Output code} / (2^{23} \cdot \text{Gain}) \quad (68)$$

2.6.5 Configuration Register Settings

For the measurement, set the register values:

- Select multiplexer settings for AINP and AINN to measure leads 2 and 3 of the RTD
- Enable the PGA, set gain to desired value
- Select data rate and digital filter settings
- Select reference input to measure R_{REF} for ratiometric measurement
- Enable the internal reference (the IDAC requires an enabled internal reference)
- Set IDAC magnitude and select IDAC1 output pin to drive lead 1 of the RTD

2.7 Two Series Two-Wire RTD Measurements, Low-Side Reference

A single IDAC may drive multiple RTDs at the same time. In this schematic, two two-wire RTDs are stacked on top of each other for measurement through different inputs of the ADC multiplexer.